

BUSINESS REQUIREMENTS DOCUMENT (BRD) v2.0

AI vs Human Content Detection System

Project Name: AI vs Human Content Detection System
Client Type: Digital Content Platform
Document Type: Business Requirements Document (BRD)
Version: 2.0 Final
Prepared By: M. Sagar Reddy
Status: Approved for Client Handover
Date: June 2026

DOCUMENT CONTROL

Version	Date	Author	Description
1.0		Project Initiation Data Science Team	Initial BRD
2.0	June 2026	M. Sagar Reddy	Final Implemented Solution & Handover Version

APPROVAL MATRIX

Role	Responsibility	Status
Client Stakeholder	Business Approval	Pending
Project Sponsor	Project Approval	Pending
Data Science Lead	Technical Approval	Completed
Client Deployment Team	Deployment Approval	Client Owned

1. EXECUTIVE SUMMARY

The rapid advancement of Generative Artificial Intelligence (GenAI) technologies has significantly increased the volume of machine-generated content being published across digital platforms. Modern Large Language Models (LLMs) are capable of producing text that closely resembles human writing, creating challenges for content moderation teams tasked with maintaining authenticity and platform trust.

The client operates a digital content platform that receives textual submissions across multiple domains, including Academic, News, and Social content. Existing moderation processes relied heavily on manual review, creating operational bottlenecks and increasing the risk of inconsistent moderation decisions.

To address these challenges, the AI vs Human Content Detection System was developed as a machine learning solution capable of automatically classifying content as either AI-generated or Human-authored.

The project progressed through multiple phases including:

- Business Analysis
- Data Validation
- Exploratory Data Analysis
- Feature Engineering
- Model Development
- Leakage Investigation
- External Validation
- Threshold Optimization
- Model Serialization
- Deployment Packaging

The final solution utilizes a hybrid Natural Language Processing architecture combining Word-Level TF-IDF features and Character-Level TF-IDF features with Logistic Regression classification.

The solution has been successfully serialized and packaged for deployment by the client engineering team.

2. BUSINESS CONTEXT

2.1 Industry Context

The widespread adoption of Generative AI platforms has transformed content creation across industries. AI-generated content now appears in:

- Academic summaries
- Research abstracts
- Blog articles
- News commentary
- Social media posts
- Product reviews
- Marketing copy

While these technologies provide operational benefits, they also create challenges related to authenticity, transparency, and content quality.

Organizations increasingly require mechanisms capable of identifying machine-generated content to preserve user trust and maintain content integrity.

2.2 Organizational Context

The client operates a digital content platform responsible for processing large volumes of user-submitted textual content.

Content originates from multiple domains:

Academic Domain

Research summaries, educational content, and analytical writing.

News Domain

Current affairs, journalism, commentary, and reporting.

Social Domain

Informal conversations, opinions, discussions, and user-generated content.

The growing volume of submissions made manual moderation increasingly difficult and costly.

2.3 Business Challenge

The organization required an automated solution capable of:

- Detecting AI-generated content.
 - Protecting authentic human-generated content.
 - Supporting moderation teams.
 - Scaling with increasing submission volume.
 - Providing deployment-ready machine learning capabilities.
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3. PROBLEM STATEMENT

The moderation team faced increasing difficulty distinguishing between AI-generated and human-authored content due to improvements in language generation technologies.

The primary business risk identified during project initiation was:

False Positive Classification

Human-authored content incorrectly classified as AI-generated.

Potential consequences include:

- Loss of creator trust
- Publication delays
- Increased moderation effort
- Reduced platform credibility
- Negative user experience

For this reason, human content protection became the most important business objective throughout the project lifecycle.

4. BUSINESS OBJECTIVES

4.1 Primary Objectives

Objective	Status
Detect AI-generated content	Achieved
Protect Human-authored content	Achieved
Reduce moderation workload	Achieved
Support automated moderation	Achieved
Deliver production-ready model	Achieved
Enable deployment packaging	Achieved

4.2 Secondary Objectives

Objective	Status
Cross-domain validation	Achieved
External validation framework	Achieved
Leakage investigation	Achieved
Model serialization	Achieved
Threshold optimization	Achieved

5. STAKEHOLDER ANALYSIS

Stakeholder	Responsibility
Content Moderation Team	Review flagged content
Data Science Team	Model development and validation
Platform Operations Team	Operational support
Client Deployment Team	Deployment and integration
Business Stakeholders	Acceptance and approval

6. PROJECT SCOPE

6.1 In Scope

The following components were included within the project scope:

Content Classification

Binary classification:

- AI Generated
- Human Written

Supported Domains

- Academic
- News
- Social

Machine Learning Activities

- Data Validation
 - Feature Engineering
 - Model Development
 - Threshold Optimization
 - External Validation
 - Serialization
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6.2 Out of Scope

The following items were excluded:

- Image moderation
 - Video moderation
 - Audio moderation
 - Multilingual detection
 - Production infrastructure setup
 - Continuous retraining pipelines
 - MLOps implementation
 - Client-side deployment execution
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7. CURRENT STATE VS FUTURE STATE

Current State

Manual moderation process:

Content Submission



Manual Review



Moderation Decision

Challenges:

- Slow
- Expensive
- Difficult to scale
- Subjective decisions

Future State

Automated workflow:

Content Submission



AI Detection Model



Prediction Score



Moderation Decision Support

Benefits:

- Faster review
- Consistent decisions
- Reduced operational workload
- Improved scalability

8. SOLUTION OVERVIEW

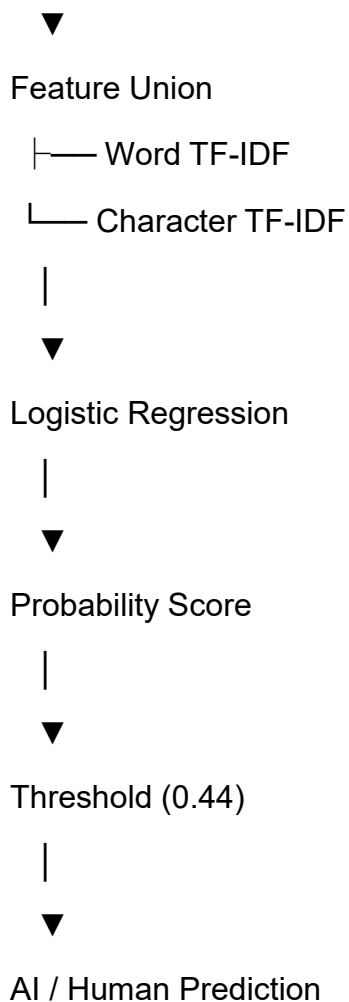
Final Architecture

Raw Text



Text Cleaning





9. DATASET OVERVIEW

Metric	Value
Total Records	2,000
Training Records	1,592
Testing Records	408
External Validation Records	39
Domains	Academic, News, Social

The dataset provided by the client was confirmed to be real-world data and not synthetic.

10. FINAL SOLUTION REALIZATION

Following multiple experiments, the final selected model was:

Word + Character TF-IDF Logistic Regression

Configuration:

- Word TF-IDF

- Unigrams
 - Bigrams
 - 5,000 Features
- Character TF-IDF
 - Character N-Grams (3–5)
 - 10,000 Features
- Logistic Regression
 - Balanced Class Weighting
- Threshold
 - 0.44

This configuration demonstrated the strongest generalization performance during external validation.

11. VALIDATION SUMMARY

Internal Validation

Metric	Score
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Accuracy	100%
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Precision	100%
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Recall	100%
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F1 Score	100%
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External Validation

Metric	Score
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Accuracy	82.05%
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Precision	81.82%
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Recall	91.67%
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F1 Score	86.47%
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External validation was used as the primary model selection criterion.

12. LEAKAGE INVESTIGATION SUMMARY

The project included a dedicated leakage investigation phase.

Activities performed:

- Duplicate content analysis
- Train/Test contamination checks
- GroupShuffleSplit implementation
- TF-IDF isolation validation
- Vocabulary leakage assessment

Outcome:

No evidence of leakage remained after implementing GroupShuffleSplit and rebuilding the feature pipeline.

13. DEPLOYMENT DELIVERABLES

The following artifacts were generated:

Artifact	Purpose
final_ai_detector.pkl	Production Model
decision_threshold.pkl	Classification Threshold
model_metadata.pkl	Model Metadata
Jupyter Notebook	Development Asset

14. RISKS AND LIMITATIONS

Risks

- Future AI models may evolve beyond current patterns.
- Domain drift may reduce accuracy over time.
- Writing styles may change.

Mitigation Strategy

- Periodic retraining
- Continuous monitoring
- Regular validation against new samples

15. PROJECT CLOSURE STATEMENT

The AI vs Human Content Detection System successfully completed all planned machine learning lifecycle activities including data preparation, feature engineering, model development, validation, leakage investigation, threshold optimization, external benchmarking, serialization, and deployment packaging.

The final solution has been approved for client handover and is ready for deployment by the client engineering team.
